

External Report

Steven Pappas and Gregory J. Matthews

2026-05-04

Our external analysis seeks to evaluate the relationship between an external country's democracy score and the likelihood of an embassy in an internal country in a given year. With the exception of official development assistance (ODA) aid, almost all modeled characteristics are independent of internal African countries. We used democracy scores from Freedom House (TOTLIB1) and the POLITY IV database (POLITY and POLITY2).

Our analysis is split into a reduced and full model. The former seeks to evaluate our model relationship without additional external country characteristics. It accounts for random effects in the internal and external countries and their interaction and the year. For the full model, we incorporated the external country's population, GNI, and ODA aid in addition to their democracy score and the same random effects. We applied a log transformation to each of these new variables to adjust for their right-skewed distribution. Finally, we applied this model framework to one to three-year lagged external democracy scores.

Reduced Model

$$Y_{ijk} \sim \text{Binomial}(n_{ijk}, p_{ijk})$$

$$\text{logit}(p_{ijk}) = \beta_0 + \beta_1 X_{ijk} + \gamma_{0i} + \gamma_{0j} + \gamma_{0ij} + \gamma_{0k} + \epsilon_{ijk}$$

$$\gamma_{0i} \sim N(0, \sigma_i^2)$$

$$\gamma_{0j} \sim N(0, \sigma_j^2)$$

$$\gamma_{0ij} \sim N(0, \sigma_{ij}^2)$$

$$\gamma_{0k} \sim N(0, \sigma_k^2)$$

Our reduced model equations are written above, where Y_{ijk} is an indicator for an embassy connection between internal country i and external country j in year k . In lagged models, the year k changes units depending on the lag. For instance, a one-year lag would produce the model equation $\text{logit}(p_{ijk}) = \beta_0 + \beta_1 X_{1ij(k-1)} + \gamma_{0i} + \gamma_{0j} + \gamma_{0ij} + \gamma_{0k} + \epsilon_{ijk}$, where $X_{1ij(k-1)}$ is the external country's democracy score in the prior year (year $k-1$).

As shown in Table 1, all external country democracy scores and their lags but the two-year lagged POLITY2 score show significance at the $\alpha = 0.05$ level. Interestingly, each model's $\hat{\beta}$ coefficients indicate that the less democratic external country j the more likely they are to have an embassy relationship with internal country i . For instance, a one-unit increase in unlagged TOTLIB1 increases the odds of an embassy relationship by 14.2 percent.

Full Model

$$Y_{ijk} \sim \text{Binomial}(n_{ijk}, p_{ijk})$$

$$\text{logit}(p_{ijk}) = \beta_0 + \beta_1 X_{1ijk} + \dots + \beta_4 X_{4ijk} + \gamma_{0i} + \gamma_{0j} + \gamma_{0ij} + \gamma_{0k} + \epsilon_{ijk}$$

$$\gamma_{0i} \sim N(0, \sigma_i^2)$$

$$\gamma_{0j} \sim N(0, \sigma_j^2)$$

$$\gamma_{0ij} \sim N(0, \sigma_{ij}^2)$$

$$\gamma_{0k} \sim N(0, \sigma_k^2)$$

Our full model equations largely follow the same setup as the reduced model with the addition of the aforementioned log-transformed external country characteristics.

As shown in Table 2, the only significant variable at the $\alpha = 0.05$ level is the unlagged POLITY score. Interestingly, despite the lack of significance among the democracy scores, their $\hat{\beta}$ coefficients display an inverse relationship to their reduced counterparts. In the unlagged POLITY score in the full model, a one-unit increase increases the odds of an embassy relationship by 1.6 percent. However, in the reduced model where it also displayed significance, a one-unit increase in the same variable decreases the odds of an embassy relationship by 6.4 percent.

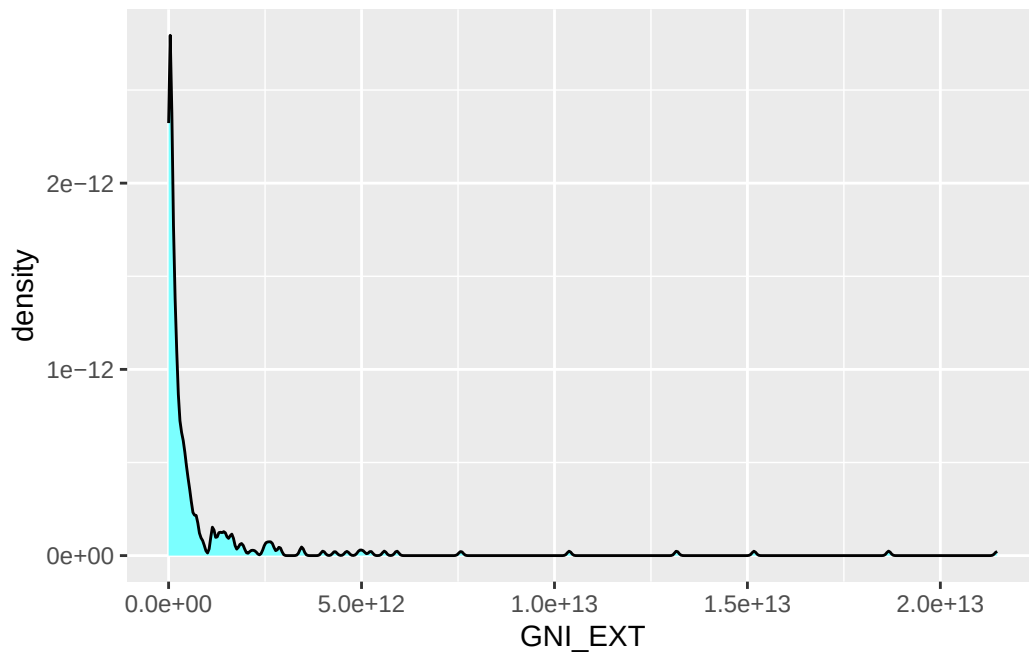
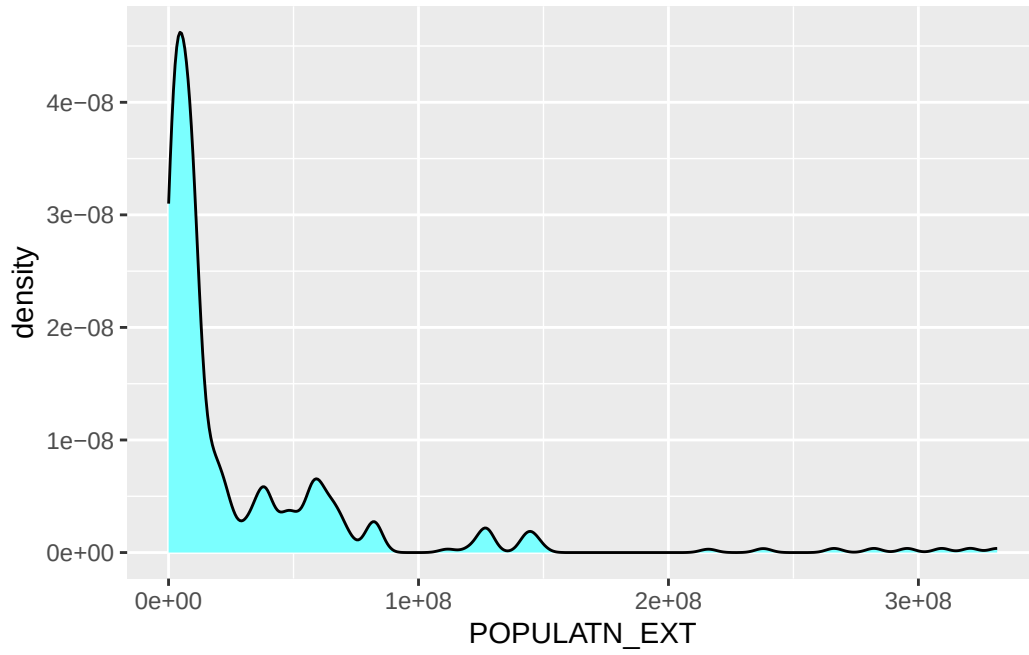


Table 1: Reduced Model Parameters for External **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	$e^{\hat{\beta}}$	p	$\hat{\beta} : 95\%CI$	$e^{\hat{\beta}} : 95\%CI$
lag0	TOTLIB1	0.185	1.203	0	0.142, 0.228	1.153, 1.256
	POLITY	-0.093	0.911	0	-0.117, -0.068	0.89, 0.934
	POLITY2	-0.090	0.914	0	-0.114, -0.066	0.892, 0.936
lag1	TOTLIB1	0.153	1.166	0	0.111, 0.196	1.117, 1.217
	POLITY	-0.098	0.907	0	-0.098, -0.097	0.907, 0.908
	POLITY2	-0.081	0.922	0	-0.104, -0.058	0.901, 0.944
lag2	TOTLIB1	0.134	1.143	0	0.093, 0.175	1.097, 1.191
	POLITY	-0.085	0.918	0	-0.107, -0.064	0.899, 0.938
	POLITY2	-0.085	0.919	0	-0.106, -0.063	0.899, 0.939
lag3	TOTLIB1	0.139	1.150	0	0.099, 0.18	1.104, 1.197
	POLITY	-0.075	0.927	0	-0.097, -0.054	0.908, 0.947
	POLITY2	-0.075	0.927	0	-0.097, -0.054	0.908, 0.947

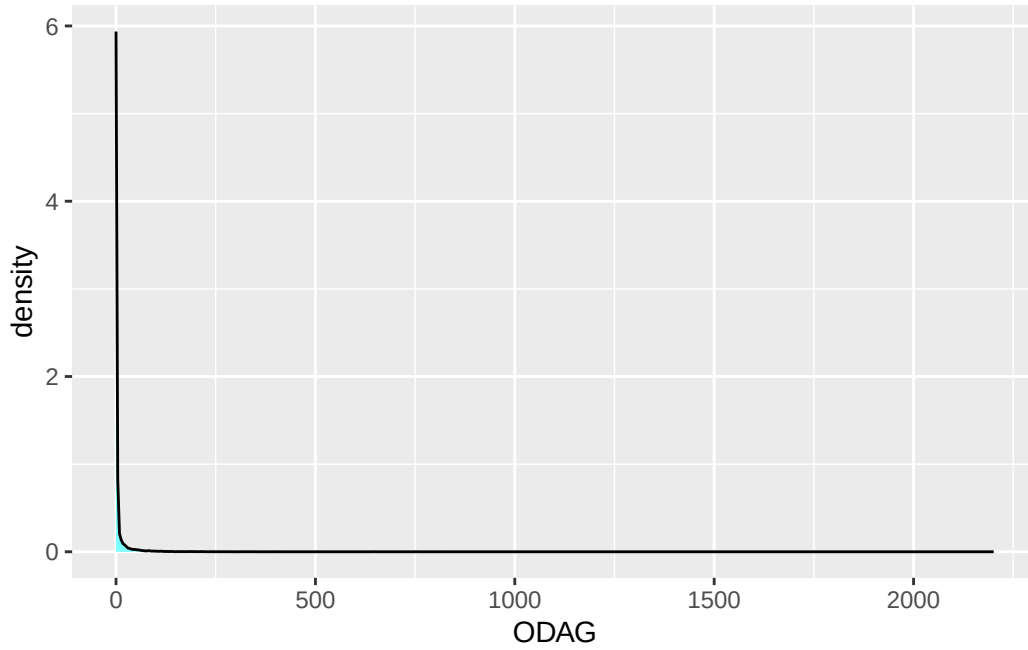


Table 2: Full Model Parameters for External **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	$e^{\hat{\beta}}$	p	$\hat{\beta} : 95\%CI$	$e^{\hat{\beta}} : 95\%CI$
lag0	TOTLIB1	0.025	1.025	0.578	-0.062, 0.111	0.94, 1.117
	POLITY	0.001	1.001	0.985	-0.065, 0.066	0.937, 1.068
	POLITY2	-0.023	0.977	0.429	-0.08, 0.034	0.923, 1.035
lag1	TOTLIB1	-0.006	0.994	0.898	-0.094, 0.082	0.91, 1.085
	POLITY	-0.032	0.969	0.255	-0.087, 0.023	0.917, 1.023
	POLITY2	-0.020	0.981	0.463	-0.072, 0.033	0.931, 1.034
lag2	TOTLIB1	-0.020	0.980	0.634	-0.102, 0.062	0.903, 1.064
	POLITY	-0.030	0.970	0.177	-0.074, 0.014	0.929, 1.014
	POLITY2	-0.030	0.971	0.186	-0.073, 0.014	0.93, 1.014
lag3	TOTLIB1	-0.004	0.996	0.000	-0.005, -0.003	0.995, 0.997
	POLITY	-0.039	0.962	0.072	-0.082, 0.003	0.921, 1.003
	POLITY2	-0.039	0.962	0.072	-0.082, 0.003	0.921, 1.003